

Slip system analysis in the cold rolling of a Ni₃Al single crystal

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Abstract. The activated slip systems were analyzed in the cold rolling of a Ni₃Al single crystal with an initial orientation of $\sim[-112](512)$, which showed an irregular rolling deformation, i.e. widening, bending, and shear deformation. A phenomenological crystal plasticity model was applied using a spectral method. The boundary condition was optimized to reproduce the actual rolling deformation, as follows. That is, the orthogonal components of the deformation gradient were given from the measured widening and reduction, and the shear components were iteratively optimized as to that the final orientation was as close to the experimental one as possible. The calculated result showed that three slip systems, **a3**, **b1**, and **d1** in the Bishop-Hill notation, were mainly activated in the irregular rolling deformation, which result was consistent to the previous observation of the slip traces [Kishida et al., Philos. Mag. 83 (2003) 3029]. The three activated systems were identical to those activated in the plane-strain condition. However, the quantitative comparison revealed that the activity of **b1** was significantly reduced in the irregular rolling deformation, while the activity of **d1** was enhanced instead. The less activity of **b1** and the enhancement of **d1** can be understood assuming a strong interaction between **a3** and **b1**. The reaction of this pair has been reported to form the superlattice intrinsic stacking fault (SISF) in Ni₃Al [Chiba and Hanada, Philos. Mag. A. 69 (1994) 751]. It is likely that the formation of the SISF, which are considered immobile in Ni₃Al, restrained the activation of **b1**, leading to the irregular rolling deformation.

Introduction

While single crystalline Ni₃Al has reasonable ductility [1], the rolling deformability significantly depends on the crystallographic orientation [2]. For example, single crystalline sheets of Ni₃Al can be cold rolled to more than 99% reduction in thickness when the rolling direction (RD) is close to $\langle 100 \rangle$ [3]. In contrast, they exhibit curving, widening, shearing, or tearing during cold rolling when the RD is close to $\langle 211 \rangle$ [2]. Such irregular rolling deformation makes further reduction difficult. A pack rolling with stainless steel sheets, which constraint can yield more regular rolling deformation, resulted in severe shear cracking. The difficulty of the regular rolling deformation suggests that some of the necessary slip systems were not activated or restrained for the RD of $\langle 211 \rangle$. It has yet to be clarified what slip systems were restrained from activating in rolling deformation.

The purpose of this study is to analyze the slip systems activated during the irregular rolling deformation. It is difficult to experimentally determine the activated slip systems from dislocation images by electron microscopy due to a large density of dislocations and stacking faults [4]. In this study, we used a computational method and calculated a shear on each slip system, which value reproduced the shape changes and the crystallographic orientation of the rolled sheet. Comparing with the slip systems necessary for a regular rolling deformation, i.e. the plane-strain deformation, we will discuss the restraining mechanism.

Computational method

Crystal plasticity model. We used a phenomenological model [5], in which a shear rate for a slip system α , $\dot{\gamma}^\alpha$, is in proportion to the n-th power of resolved stress, τ^α , as follows:

$$\dot{\gamma}^\alpha = \dot{\gamma}_0 \left| \frac{\tau^\alpha}{S^\alpha} \right|^n \text{sgn}(\tau^\alpha). \quad (1)$$

S^α is a resistance, modeling a strain hardening, that is assumed to increase from the initial value, S_0 , with shear deformation. The increase rate, $\dot{S}^\alpha$, is described, as follows:

$$\dot{S}^\alpha = h_0 (1 - S^\alpha / S_\infty)^w \sum_\beta h_{\alpha\beta} \dot{\gamma}^\beta, \quad (2)$$

where $h_{\alpha\beta}$ represents the strength of each slip-slip interaction and was set as 1.4 for non-coplanar interaction, higher than for the coplanar one in order to take a latent hardening effect [6]. The value of each parameter except for the slip-slip interaction was chosen to reproduce a tensile stress-strain curve of a Ni₃Al single crystal. Table 1 lists the values of these parameters as well as the elastic constants. As a solver for the mechanical boundary value problem, we applied a spectral method using the fast Fourier transform, which method is more efficient than a widely used finite element method [7]. We used the Düsseldorf Advanced Materials Simulation Kit (DAMASK) [8].

Table 1. Parameters for the phenomenological plasticity model used in this study.

$\dot{\gamma}_0$	n	h_0	S_0	S_∞	w	Coplanar $h_{\alpha\beta}$	Non-coplanar $h_{\alpha\beta}$	C11	C12	C44
0.002	50	210 MPa	30 MPa	5000 MPa	1	1	1.4	223 GPa	148 GPa	125 GPa

Optimization of the boundary condition. The boundary condition was given in a form of an average deformation gradient, $\bar{F}_{ij} = \partial x_i / \partial X_j$, which consists of nine components. The orthogonal components, $\bar{F}_{22}$, $\bar{F}_{33}$, and $\bar{F}_{11}$, were given from the widening and reduction of the rolled sheet and determined from the constant volume condition, respectively. Assuming that the ND and the RD keep their initial directions, three shear components, $\bar{F}_{21}$, $\bar{F}_{32}$, and $\bar{F}_{31}$, are set as Zero.

There are three remaining shear components, $\bar{F}_{12}$, $\bar{F}_{13}$ and $\bar{F}_{23}$. We iteratively optimized the set of these components as to that the final orientation is as close to the measured one as possible, using the simplex method [9].

Calculation details. The periodic volume element was taken as a cube of 1 x 1 x 1 m³, discretized by 4 x 4 x 4 for the optimization process of the boundary condition or by 8 x 8 x 8 voxels for computing the results used for analysis. The obtained results were irrelevant to the resolution of discretization, which is one of the merits of the spectral method.

A cold-rolled specimen. From the previous study [2], we chose the specimen #B2-41-1 as a representative example, which shows a widening, a bending, and a shear deformation. The initial orientation was [-0.47, 0.4, 0.79] (0.88, 0.21, 0.42). The final orientation was determined from the {220} and {111} pole figures measured in the rolled sheet. The shape change and the orientation of the rolled sheet are summarized in the last line of Table 2.

Results and discussions

The boundary condition was optimized within 50 iterations, and two different initials yielded the identical solution, supporting the validity of the optimization process. The obtained boundary

condition (listed in Table 2) reproduces the lattice rotation and shape change satisfactorily. Firstly, the final orientation is virtually the same as that experimentally obtained; in fact, the deviation is 0.0° . Secondly, the value of the xy shear component, -0.55, is reasonably close to the measured value, -0.4, which was estimated from the inclination angle of the sheet edge.

Table 2. The boundary conditions and the final orientations.

Boundary condition	Shape change of rolled sheet*						Final orientation			
	xx	yy	zz	xy	yz	xz	phi1	Phi	phi2	Deviation
Optimized	1.33	1.19	0.63	-0.55	-0.24	0.06	129.9	60.0	78.0	0.0
Plane strain	1.58	1.00	0.63	0.00	0.00	0.00	132.3	71.2	70.1	13.4
Compression	1.00	1.59	0.63	-0.03	0.17	0.04	110.7	57.4	99.5	20.2
Experimental	-	1.19	0.63	-0.4	-	-	129.9	60.0	78.0	

Fig. 1 (a) shows the lattice rotation through the deformation under the optimized boundary condition. The orientation rotates from the initial one (shown by the darker solid circles) toward the final one (the crosses). The ND tends to rotate toward [101] (the rectangle #2), which tendency is frequently observed in the rolling of Ni_3Al single crystals [2,3]. Fig. 1 (b) plots the compression stress with respect to the ND as a function of the reduction, together with the results calculated under the plane-strain and compression conditions. In the early stage to 5% reduction, the strain-hardening curve follows that for the compression condition, and after showing a sharp increase, it follows the curve for the plane-strain condition. The final stress is almost the same or slightly higher than that for the plane-strain condition. This suggests that the irregular rolling deformation is not necessarily favorable to reduce the deformation stress.

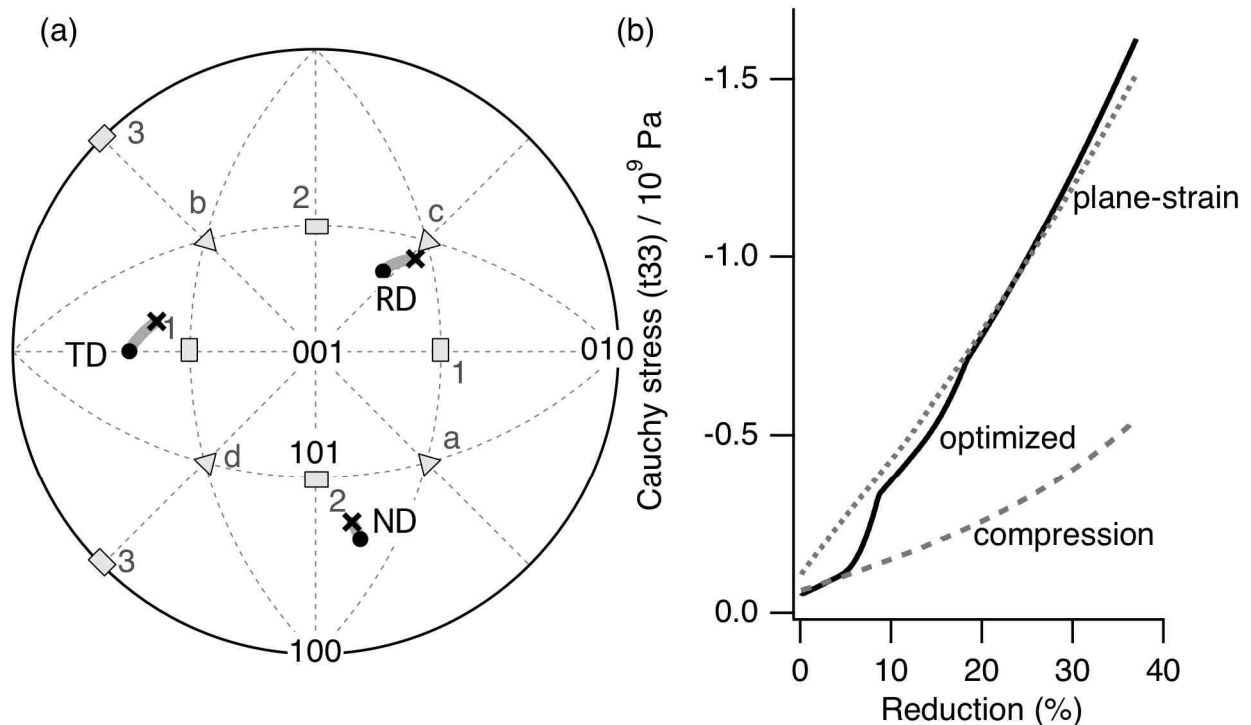


Figure 1. Lattice rotation (a) and Cauchy stress along (b) the ND through the deformation under the optimized condition. The solid circle and cross in (a) represent the initial and final orientations experimentally measured, respectively, and the triangle and rectangle represent the slip plane and direction labeled according to the Bishop-Hill notation, respectively.

Table 3 summarizes the contribution of each slip system to the total shear; each slip system is labeled the combination of the slip plane (**a**, **b**, **c**, or **d**) and the slip direction (**1**, **2**, or **3**) according to the Bishop-Hill notation [10], as shown in 001 standard stereographic projection of Fig. 1 (a). The shear was calculated by integrating the shear rate recorded per second and the contribution is the ratio of the absolute value of the shear to the total shear that is the sum for all the slip systems.

In the optimized boundary condition, three slip systems, **a3**, **b1**, and **d1**, are mainly activated. In detail, the shear on **a3** and **d1** is much larger than that on **b1**. These results are consistent with our previous slip-trace analysis [2], as follows: the slip deformation occurred on the planes, **a**, **b**, and **d**; and the slip traces were more densely observed for **a** and **d**, while the trace for **b** was less observed. Thus, the analysis on the major slip systems reproduces the actual rolling deformation well. Besides of these major contributions, there are several slip systems whose contribution is less than 10% of the total shear, but it seems difficult to analyze the activation of these minor slip systems in detail due to the small contribution. Therefore, it is more reasonable to confine the analysis to the major slip systems whose contribution is over 10% (underlined in Table 3).

Table 3. The contribution of each slip system for various boundary conditions

Boundary condition	Contribution of each slip system to the total shear											
	a1	a2	a3	b1	b2	b3	c1	c2	c3	d1	d2	d3
Optimized	0.05	0.02	<u>0.41</u>	<u>0.12</u>	0.00	0.02	0.00	0.00	0.01	<u>0.31</u>	0.00	0.06
Plane strain	0.00	0.06	<u>0.32</u>	<u>0.40</u>	0.01	0.00	0.00	0.00	0.00	<u>0.19</u>	0.00	0.01
Compression	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	<u>1.00</u>

The three major slip systems are identical to those activated in the plane-strain condition (Table 3). However, the contribution of **b1** is significantly reduced in the actual rolling deformation, and that of **d1** is increased instead. These changes lead to the irregular deformation far from the plane-strain deformation. In addition, it turned out that the slip systems are completely different from that for the compression condition, meaning that the slip systems in the rolling deformation were activated in a different way from that under the compressive stress state.

We then discuss possible reasons of the less activation of **b1** and the enhancement of **d1**. Firstly, we test the idea that the amount of the shear on each slip system is determined by the magnitude of the resolved shear stress under an external stress. In the rolling deformation the external stress is approximated by a biaxial plane stress state, in which the magnitude is identical for the compressive stress along the ND and tensile stress along the RD (see the overview by Hirsch and Lücke [11]). According to this approximated rolling stress state, we calculated the resolved shear stress for each slip system as shown in Fig. 2; here, the magnitude of the stress is set unity. Among the three slip systems, the magnitude decreases in the order of **a3**, **b1**, and **d1** until 30% reduction. This order is inconsistent with the order of the magnitude in the actual rolling deformation, except for the agreement that **a3** is the largest. Thus, the magnitude of the resolved shear stress under the approximated rolling stress state cannot explain the less activation of **b1** and the enhancement of **d1**.

We next consider the effect of the interaction between slip systems. A multiple slip raises a stress necessary for slip deformation, known as the latent hardening [12], which yields a less activation of a latent slip system if the interaction with a primary system is strong. Among the two possible interaction involving **b1**, not the interaction with **d1** but the interaction with **a3** should be responsible for the less activation of **b1**, considering the enhancement of **d1**. Also, we should assume the **a3-b1** interaction is much stronger than the **a3-d1** interaction. In summary, the strong **a3-b1** interaction and the relatively weaker **a3-d1** interaction are the necessary assumptions to explain the less activation of **b1** and the enhancement of **d1**.

In the fcc structure, based on which the ordered structure of Ni_3Al is formed, the interactions are classified into six, as follows: self, coplanar, collinear, Hirth lock, glissile junction (GJ), and Lomer Lock interactions [13]. The **a3-b1** and **a3-d1** interactions correspond to the GJ and Lomer types in the fcc structure, respectively. It seems unlikely to assume the strong GJ interaction and the weaker Lomer-lock interaction from the analogy with fcc metals, in which the Lomer-lock interaction is stronger [12] or comparable to [14] the GJ interaction.

In Ni_3Al , however, the reaction of the GJ pair yields a different defect from the glissile junction in the fcc metals. It is a superlattice intrinsic stacking fault (SISF) bounded by $1/3\langle 211 \rangle$ super Shockley partials. The reaction mechanism has been proposed by Kear et al. [15] and proved with the TEM analysis by Chiba and Hanada [16]. Our previous TEM observation also confirmed the formation of SISF in a cold-rolled Ni_3Al single crystal [4]. The SISF of Ni_3Al is considered to be virtually immobile due to the large Burger's vector of the super Shockley partials. Thus, the reaction between the GJ pair, **a3** and **b1**, can yield the immobile defect, which should suppress their simultaneous activation. It is likely that due to the highest resolved shear stress under the approximated rolling stress state (Fig. 2), **a3** was activated as a primary system, reducing the activation of **b1** with the formation of the immobile SISF.

Regarding the Lomer-lock interaction of **a3** and **d1**, we could not find any reports on the dislocation reaction of the Lomer pair in Ni_3Al , though the Lomer configuration was observed in a single slip of $\langle 110 \rangle \{100\}$ at high temperatures [17]. It is out of scope in this paper to discuss the possible reaction of the Lomer pair in Ni_3Al , but the present result suggests that the interaction of the Lomer pair is not so strong as to reduce the activation of **d1**.

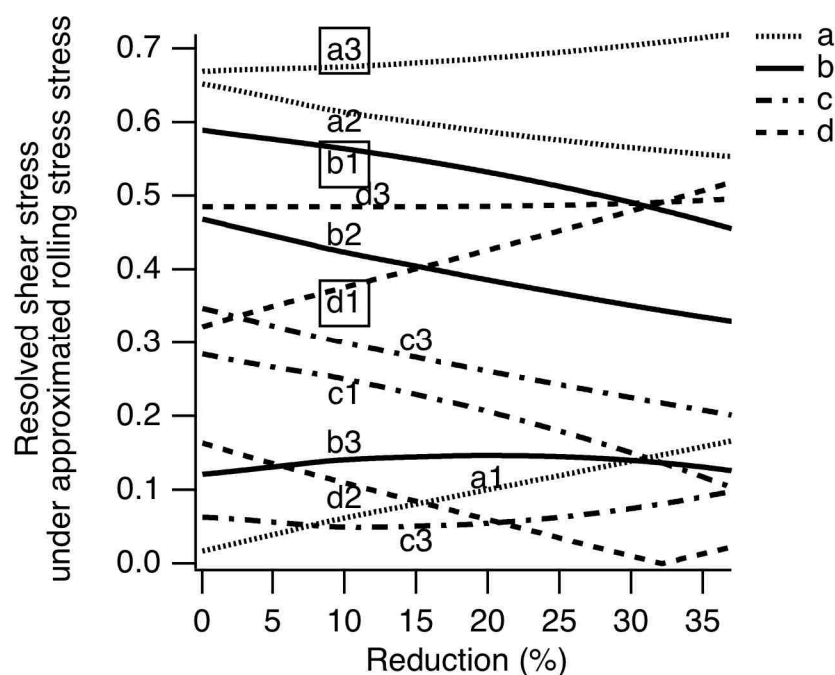


Figure 2. Resolved shear stress under the approximated rolling biaxial stress state.

Summary

The activated slip systems were quantitatively analyzed in the rolling deformation of the Ni_3Al single-crystalline sheet, using the computational method. The optimized boundary condition was iteratively determined to reproduce the shape change and the final orientation of the rolled sheet. The slip system analysis revealed that the three slip systems, **a3**, **b1**, and **d1**, were activated in the present rolled sheet. From the comparison with that for the plane-strain condition, we found that **b1** was less activated, while the slip on **d1** was enhanced. The less activation of **b1** and the enhancement of **d1** were inconsistent to the order in the magnitude of the resolved shear stress under the approximated rolling stress state. These results suggested the strong interaction between

a3 and **b1**, which is classified into the glissile-junction pair in the fcc structure. As a possible mechanism of the strong interaction, the formation of the immobile SISF was proposed.

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